BridgeSTM：A Low-Cost and Easy-to-Maintain Scanning Tunneling Microscope with a Flexure-Hinge Coarse-Approach Mechanism

Introduction

The Scanning Tunneling Microscope (STM) is a crucial experimental instrument that achieves atomic-scale surface imaging and material characterization by measuring the tunneling current between a metal probe and a conductive sample. It finds extensive applications in condensed matter physics, materials science, and nanotechnology research. However, commercial STM systems are typically expensive, structurally complex, and demand high standards for operating environments and maintenance levels. These factors have somewhat limited their adoption and application in resource-constrained research settings and entry-level scientific platforms.

In recent years, research efforts, along with related patents and open-source designs, have explored simplified mechanical structures and low-cost STM implementation strategies. These projects demonstrate that low-cost nanoscale atomic imaging can be achieved through self-built STMs featuring independently constructed circuits, customized mechanical designs, and simplified piezoelectric scanners. Most projects have successfully scanned the standard STM sample HOPG and obtained high-quality atomic images. This preliminary research provides a crucial foundation for the feasibility of constructing structurally simplified STM systems. However, existing designs still commonly face the following issues: First, coarse-approach mechanisms are structurally complex, requiring significant time for maintenance or relying on inefficient manual operation. Second, the cost of building a system remains relatively high for most departments with limited budgets. Third, most STM designs, particularly mechanical components, experience frequent friction during operation, leading to noticeable vibrations and thermal expansion effects at the microscopic level.

This research builds upon my first STM prototype system (sixth-generation design). The current system has achieved: (1) A coarse-approach mechanism based on a bridge-type flexible hinge structure, enabling the probe to gradually approach the sample surface; (2) A 1/4-quadrant piezoelectric scanning actuator providing micro-displacement drive in a single direction; (3) Application of a controlled 0-3V bias voltage to the metal sample stage. However, the system lacks completed tunnel current detection and closed-loop feedback circuitry, and critical experimental data (e.g., I–Z characteristic curves) remain unobtained. Its stability and repeatability require systematic evaluation.

This project aims to integrate and experimentally characterize the flexible hinge coarse-drive mechanism and embedded electronic control system through engineering integration and experimental characterization, building upon prior research and relevant design concepts while incorporating numerous patents and solutions. The goal is to upgrade the existing engineering prototype into an STM system with basic experimental capabilities. The research focuses on constructing a simplified, maintainable STM architecture with stable operational capabilities. Through experimental measurements, it will validate the feasibility of this architecture for tunneling current detection and displacement control, thereby providing a reference for designing simplified scanning probe instrument systems.

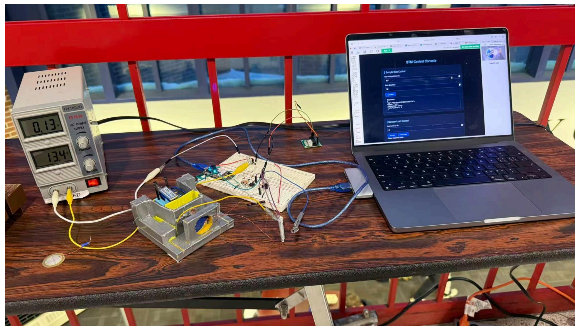

Fig1. 1st STM prototype I made

Hypothesis

This project builds upon the first STM prototype to further develop and optimize the coarse approach mechanism with flexible hinges and the embedded electronic control system. It aims to achieve a simplified and easily maintainable scanning tunneling microscope (STM) system that maintains the mechanical stability and positioning accuracy required for tunneling current measurements while simplifying the STM structure.

Method

Mechanical Structure: Design and optimize a coarse approach mechanism based on flexible hinges. Building upon the patented “MOTION REDUCING FLEXURE STRUCTURE” and referencing its engineering application in JWTS telescopes, further refine the prototype's existing flexible hinge structure. This optimization considers multiple factors including material stiffness, thermal expansion coefficient, and cost. Simultaneously, rapid iteration of mechanical prototypes was achieved through 3D printing/metal 3D printing technologies.

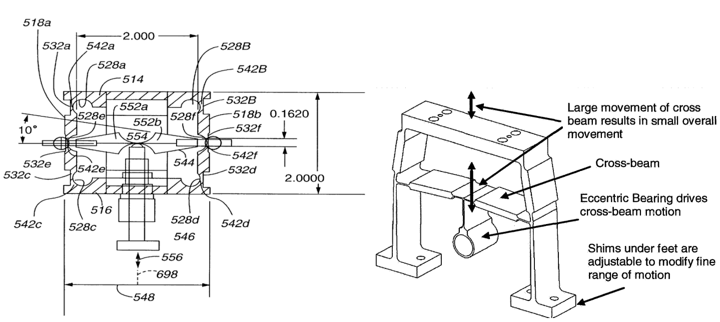

Fig 2. The Bridge-Type Fluxer & it’s Engineering application on JWTS

Control System and Circuitry: I will attempt to build the STM's control system visualization interface using Python with PyQT5 and the STM's embedded system using C++. Additionally, to achieve four-quadrant control of the piezoelectric scanner, I will employ Apex Microtechnology PA85/PA107 high-voltage operational amplifiers connected to MCP4822/AD5724 DACs for continuous high-voltage drive (0-150V) to precisely control the scanner's displacement. Furthermore, since STMs typically require detecting currents at the nanoampere level, I will employ a Femto DLPCA-200 transimpedance amplifier to amplify the tunneling current and reduce noise interference affecting scanning results. Additionally, I will replace the MCU from Photon2IoT with Tenssy4.1 to avoid noise generated by Photon's built-in Wi-Fi module. This is why I believe Photon is unsuitable as the MCU: The inherent interference from the Photon module prevents the STM from operating in fast scanning mode, forcing it to adapt to slow scanning—significantly impacting the STM's ability to achieve nanometer-level scanning resolution. In contrast, the Tenssy is highly suitable as the new STM's MCU because it allows all necessary components to be externalized and generates minimal noise.

Experimental Verification: To verify whether the piezoelectric scanner of the STM achieves nanometer-level resolution, I will attempt to construct a simple experimental setup based on the optical lever principle: The bottom of the piezoelectric scanner will have its probe base removed and a small mirror fixed in its place. A laser beam will then be directed onto the mirror surface. The minute displacement of the reflected light spot into the lens will be amplified. Subsequently, a Python program I wrote will analyze the pixel movement of the light spot to calculate whether the scanner's accuracy meets the required standard. To verify the functionality of the assembled STM, I will scan the prototype using the standard calibration sample for most STMs: graphite obtained from pyrolytic graphite (HOPG). I will obtain an I-V curve. If the curve matches those obtained by most researchers, it will demonstrate that the STM has achieved preliminary scanning capability. This will enable subsequent image scanning procedures and atomic-level imaging.Advances in AI technology will significantly enhance the efficiency of these projects.

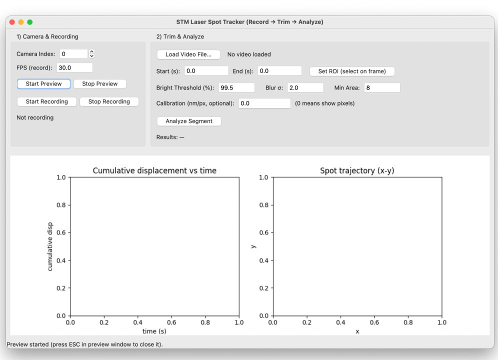

Fig.3 Python program I wrote for analyze the pixel movement of the light spot

Outcome

Upon completion, this project is expected to achieve a structurally simplified STM based on a coarse-approach mechanism with flexible hinges, implement I-V curve (tunneling current detection) functionality, and provide design insights for low-cost, easy-to-assemble, and highly maintainable STMs.This project will provide a validated low-cost STM prototype and design framework for simplified scanning probe instruments.This project will provide a validated low-cost STM prototype and design framework for simplified scanning probe instruments.

Reference

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